Bank Marketing Campaign Analysis

Which customers subscribe to a term deposit, and how should the bank prioritise its calls?

Data: direct phone-marketing campaigns of a Portuguese bank (2008-2010), 41,188 contacts, 21 variables (UCI Machine Learning Repository, bank-additional-full). Prepared by Himanshu Prakash, 2026.

Executive summary

Only **11.3%** of the 41,176 unique contacts subscribed to a term deposit. This report identifies who converts, groups customers into actionable segments, and builds a pre-call scoring model so that the call centre can work its list in order of likelihood to say yes.

- **History with the bank is the strongest signal.** Customers who accepted an earlier campaign converted at **65.1%**, versus 8.8% for customers never contacted before (5.8x lift). They are only 3.3% of the base.
- **Age and occupation matter.** Customers aged 65+ convert at 47.3% and under-25s at 24.0%; students (31.4%) and retirees (25.3%) lead by job. Blue-collar workers convert at 6.9%.
- **Channel and persistence.** Cellular contacts convert at 14.7% vs 5.2% on fixed telephone, and success falls from 13.0% on the first call to 3.6% by the tenth.
- **A pre-call model ranks customers well.** The selected gradient boosting model reaches a test ROC-AUC of 0.81; the top 10% of scores convert at 53% (4.7x lift) and the top 20% captures 67% of all subscribers.
- **Illustrative business case.** Under assumed costs (EUR 10 per call, EUR 100 per deposit), calling only the top-scored ~19% of customers raises profit per call from about EUR 1 to EUR 29.
- **Important caveat.** On a time-forward test (train on earlier calls, test on the latest 20%) AUC falls to 0.64-0.67, so the headline figures are optimistic and should be validated on live calls before scaling.

Recommended actions: (1) rank every customer before dialling; (2) call past responders first; (3) prefer cellular numbers and cap repeat attempts; (4) pilot tailored offers for retirees, students and under-25s with a holdout group to measure real uplift.

1. Data and approach

Each row is one customer contact in a phone campaign. The target is whether the customer subscribed to a term deposit. Variables fall into four groups: customer profile (age, job, marital status, education, credit default, loans), current-campaign details (channel, month, weekday, number of calls, call duration), history from earlier campaigns, and five macro-economic indicators.

The work is organised in five notebooks (data understanding, cleaning, exploratory analysis, segmentation, prediction); every table and figure in this report is produced by them and stored in the *outputs* folder.

Data quality and cleaning

- No explicit missing values, but **hidden missingness**: 'unknown' appears in default (20.9% of rows), education (4.2%), housing and loan (2.4%), job (0.8%). It was kept as its own category.
- **12 exact duplicates** removed, leaving 41,176 rows.
- pdays = 999 (96.3% of rows) means 'never contacted'; converted to a flag.
- Number of calls capped at 10 for modelling; eight engineered features added (age band, job group, loan flag, etc.).
- **Call duration is excluded from prediction.** It is only known after the call and would leak the answer (AUC 0.95 with it versus 0.81 without).

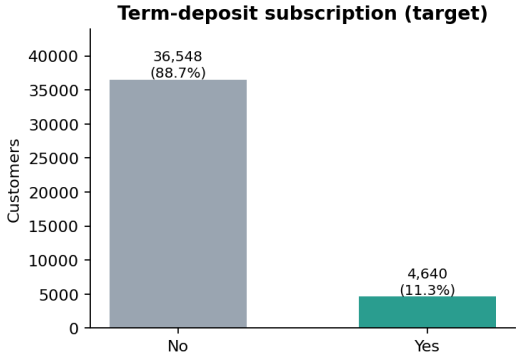


Figure 1. The target is imbalanced: about 1 in 9 contacts subscribed, so ranking metrics are used instead of accuracy.

2. What drives conversion?

For each driver we compare the segment's conversion rate with the 11.3% baseline. Lift is the ratio of the two.

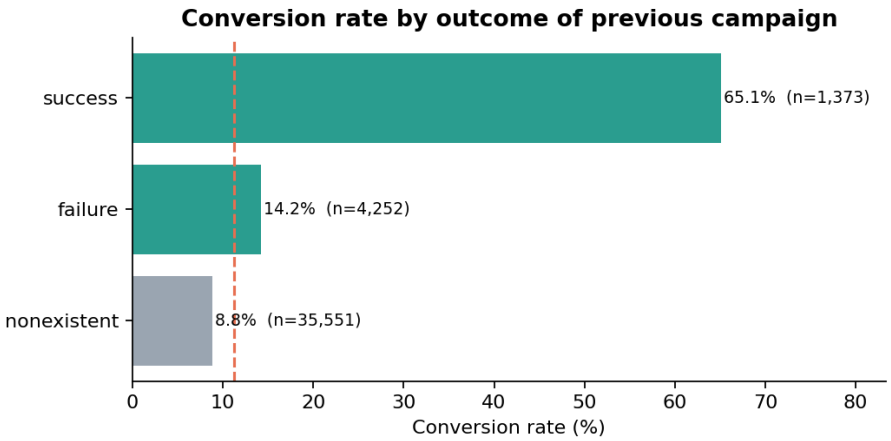


Figure 2. Outcome of the previous campaign. Past success converts at 65.1%.

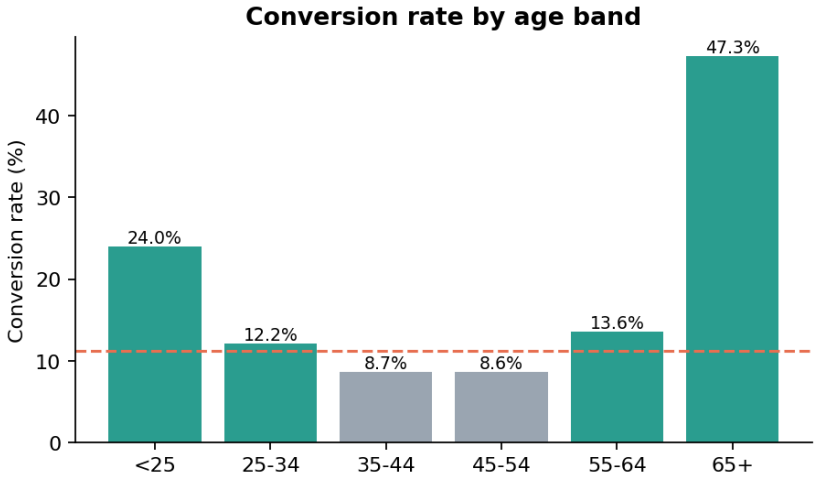


Figure 3. Conversion is U-shaped in age: the youngest and oldest customers respond best.

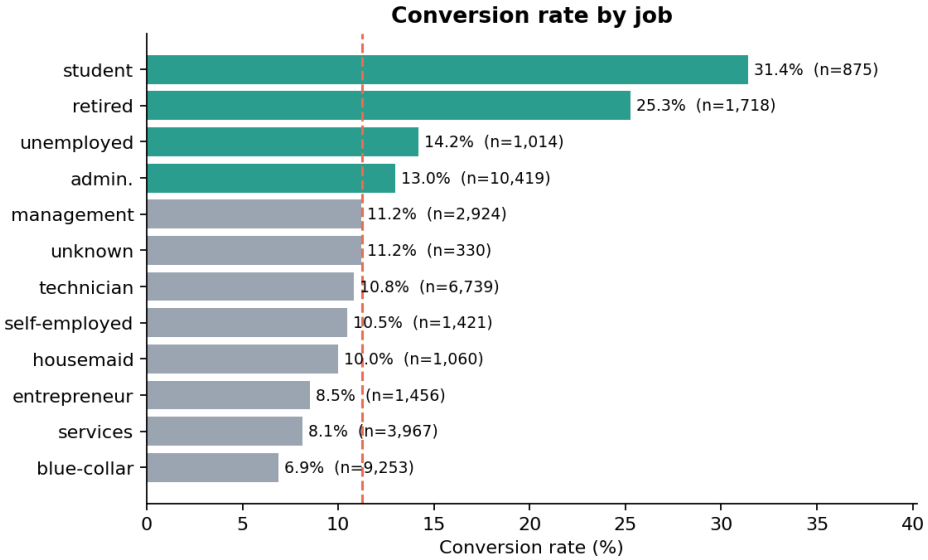


Figure 4. Students and retirees convert at 2-3x the average.

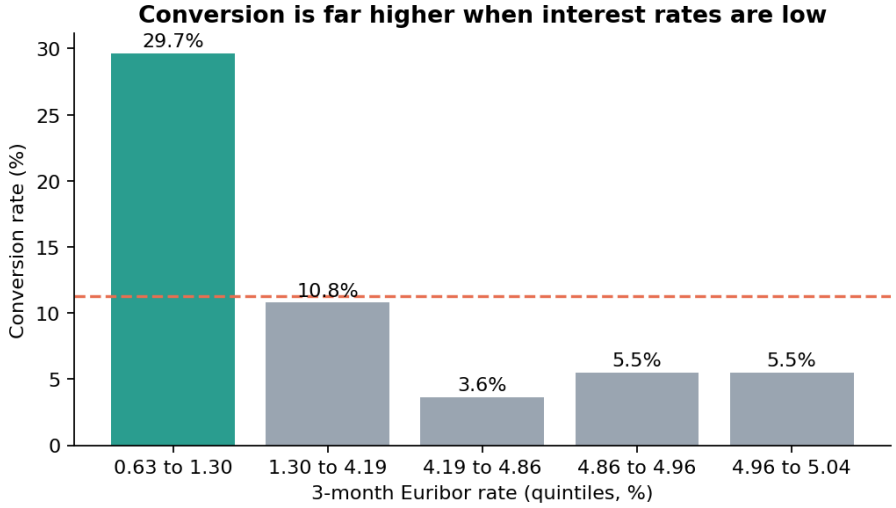


Figure 5. Conversion is highest when the 3-month Euribor rate is low (29.6% in the lowest quintile).

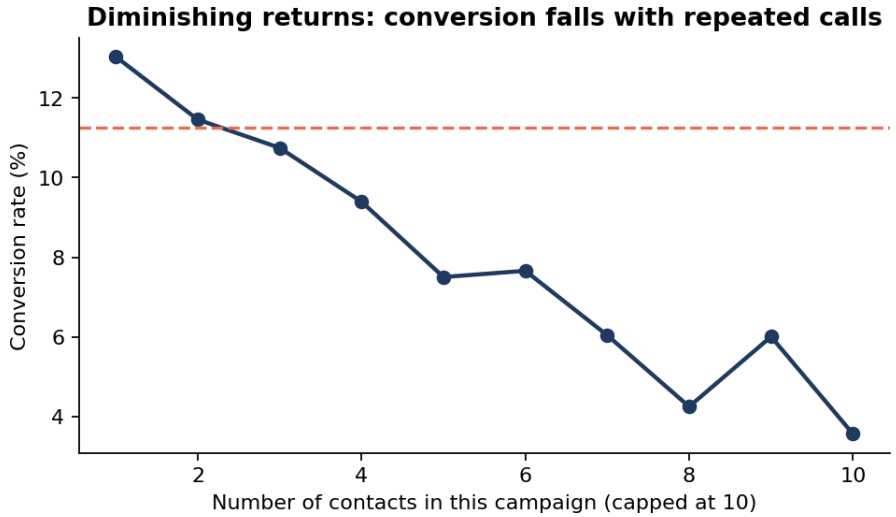


Figure 6. Diminishing returns: each additional call to the same person is less likely to succeed.

Month effects: May took a third of all calls (13,767) and converted at only 6.4%, while March, September, October and December converted at 44-51% on about 2,000 calls combined, probably warmer lists. Education, marital status and loans move conversion by under 4 percentage points and are weak targeting levers.

Interpretation note: these are associations in observational data. Month and economic effects are intertwined with who the bank chose to call and when, and should not be read as causal.

3. Customer segments

Two views were built. The rule-based view crosses age band with job group; the data-driven view applies K-Means to attributes known before the call (profile and past-campaign history). Four clusters were chosen using the highest silhouette score (0.18); this value is modest, so the segments are overlapping profiles rather than sharply separated groups.

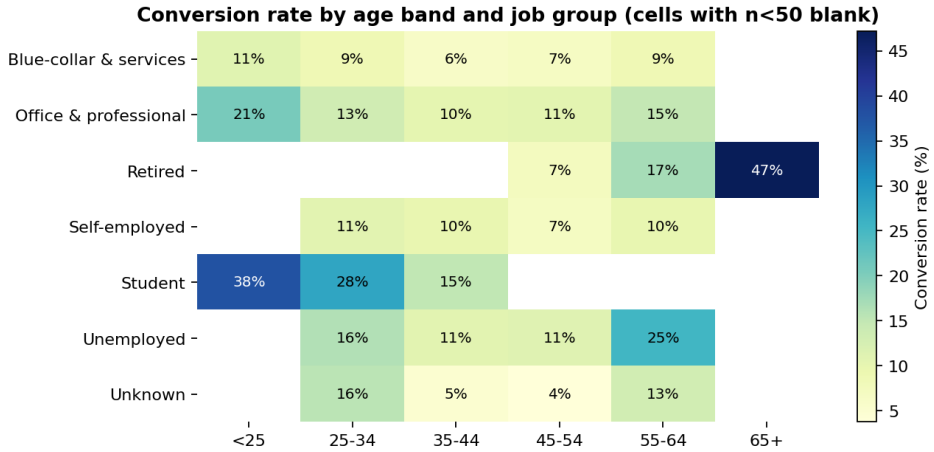


Figure 7. Best pockets: retired 65+ (47.2%), under-25 students (37.8%), 25-34 students (27.9%).

Segment	Customers	Conversion	Lift	% of customers	% of subscribers	Priority
Past-campaign responders	1,515	63.8%	5.67x	3.7%	20.8%	Protect & prioritise
Past-campaign non-responders	4,079	12.7%	1.13x	9.9%	11.2%	Maintain
Young first-time prospects	20,990	9.3%	0.83x	51.0%	42.3%	Reduce / re-target
Mature first-time prospects	14,592	8.2%	0.73x	35.4%	25.7%	Reduce / re-target

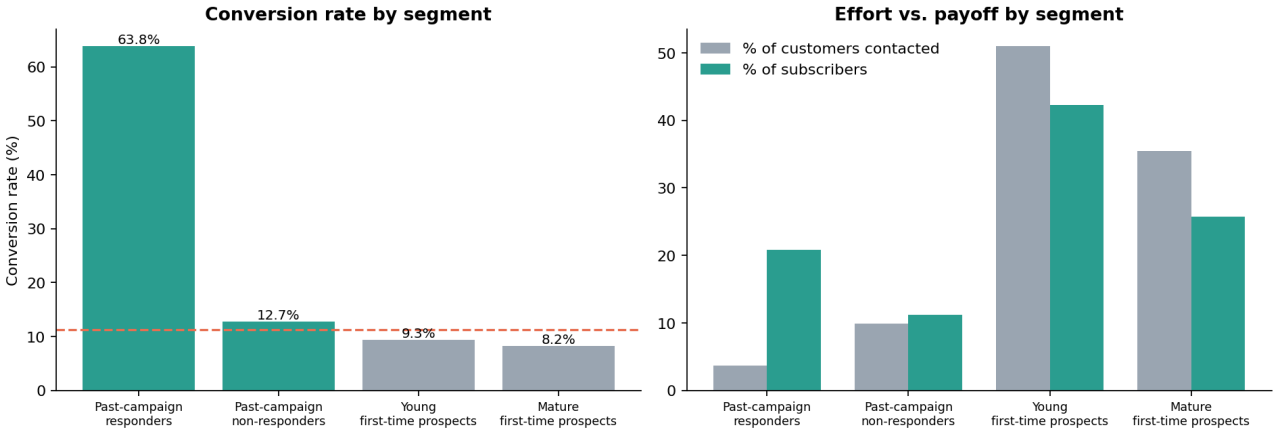


Figure 8. Past responders are 3.7% of customers but 20.8% of all subscribers.

First-time prospects make up 86% of the base and convert at only 8-9%, so the main value comes from **ranking within this large group**, which is what the predictive model does.

4. Predicting who will subscribe

Three models were compared on identical features (customer profile, contact channel, timing, past-campaign history and macro-economic indicators; call duration excluded) using an 80/20 stratified split and 5-fold cross-validation on the training portion.

Model	CV AUC (mean)	Test ROC-AUC	Test PR-AUC
Logistic Regression	0.790	0.800	0.446
Random Forest	0.798	0.811	0.483
Gradient Boosting	0.801	0.812	0.493

The baseline PR-AUC for random ranking equals the 11.3% positive rate. Gradient boosting was selected, although the differences between models are small.

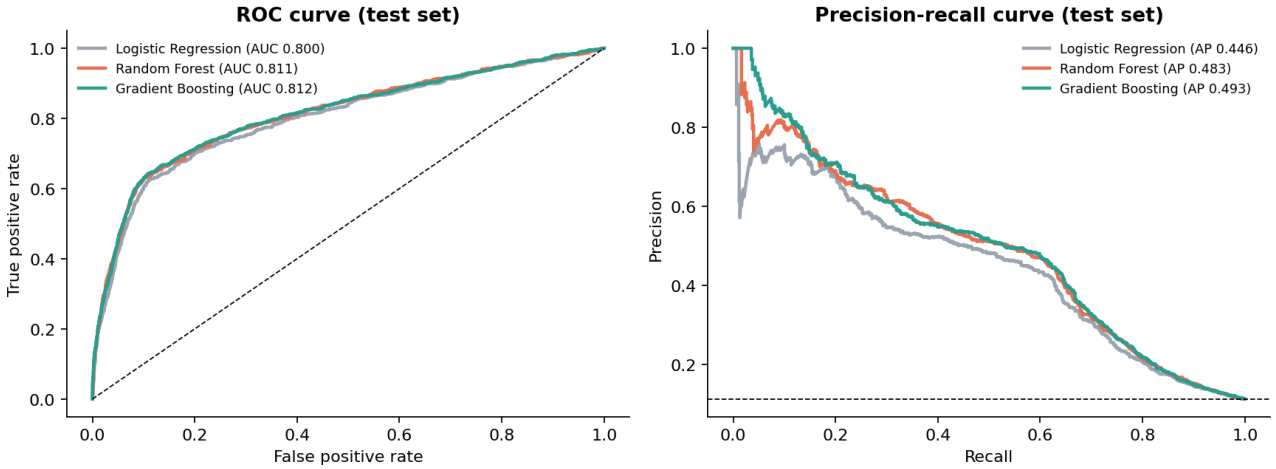


Figure 9. ROC and precision-recall curves on the held-out test set.

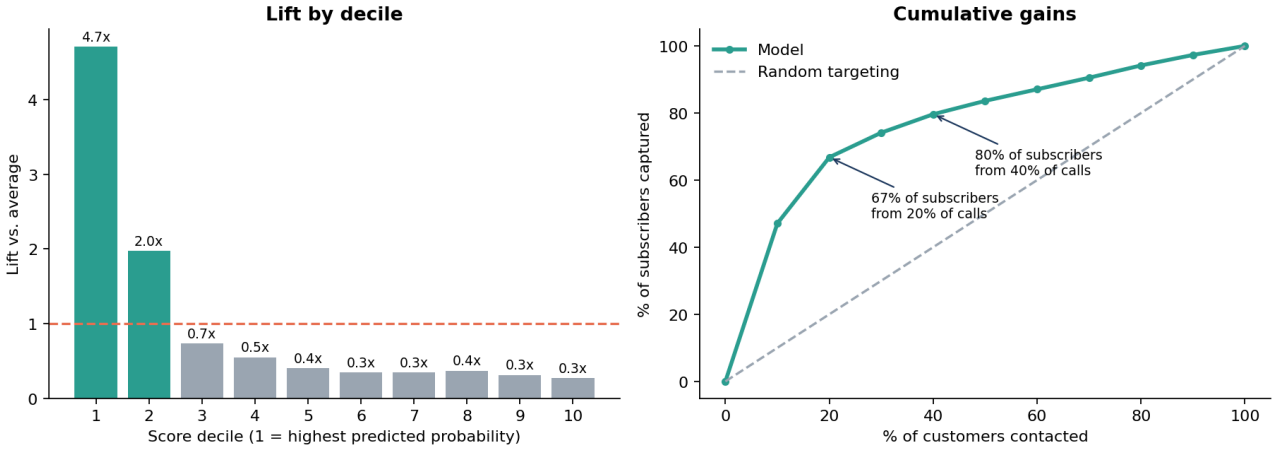


Figure 10. The top decile converts at 4.7x the average; the top 20% of customers contains 67% of subscribers.

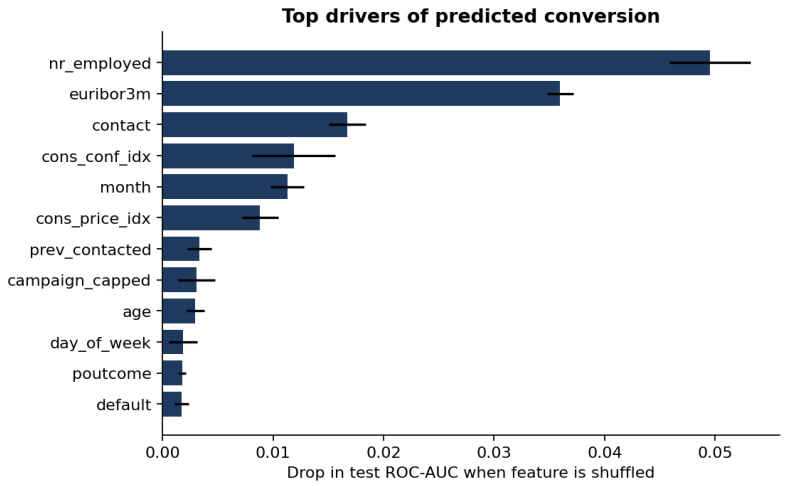


Figure 11. Permutation importance: economic indicators lead, followed by contact channel and month.

Score bands for the call centre

Band	Customers (test)	Conversion	Lift	Recommended action
A (top 10%)	823	53.1%	4.7x	Call first, senior agents
B (next 20%)	1,647	15.2%	1.4x	Call, standard script
C (next 30%)	2,471	4.9%	0.4x	Call only if capacity allows
D (bottom 40%)	3,295	3.6%	0.3x	Do not call / use low-cost channel

5. Business impact (illustrative)

To translate ranking into money, assume each call costs EUR 10 and each subscription is worth EUR 100 in net margin, so the break-even conversion rate is 10%. Calling customers in score order and stopping where profit peaks gives the comparison below (scaled to the full 41,176-customer base).

Strategy	Calls	Subscribers captured	Profit (EUR)	Profit per call (EUR)
Call everyone (baseline)	41,176 (100%)	100%	52,195	1.3
Call top-scored customers only (model cut-off)	7,944 (19%)	66%	229,028	28.8

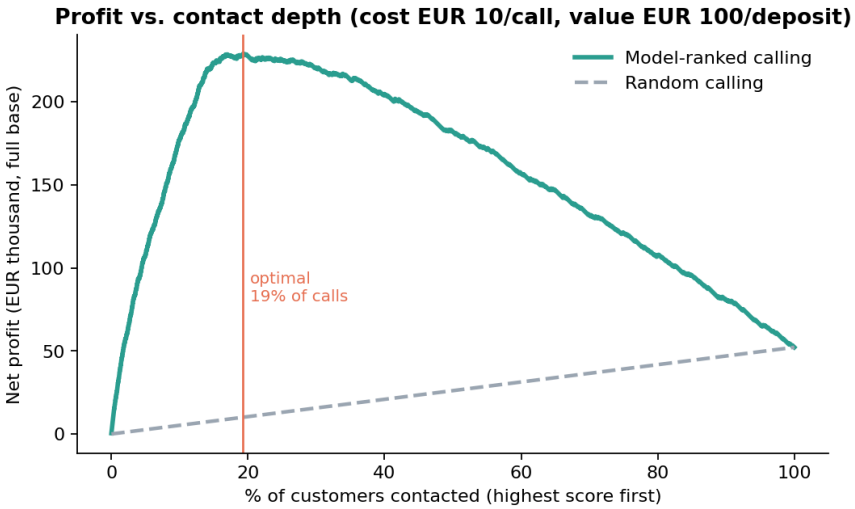


Figure 12. Profit by contact depth. Both the cost and value figures are assumptions to be replaced with the bank's actuals.

The cut-off was chosen on the test data, so real-world gains will be smaller; the ordering of the strategies, not the euro amounts, is the robust conclusion.

6. Robustness and limitations

Two extra checks probe how much the headline result depends on the evaluation design.

Check	Test ROC-AUC
Random split, all features (selected model)	0.812
Random split, NO macro features	0.781
Chronological split, all features	0.638
Chronological split, NO macro features	0.667

- **Time-forward performance is much lower.** Training on the earliest 80% of calls and testing on the latest 20% gives AUC 0.64-0.67, versus 0.81 for a random split. The random-split figure is therefore optimistic.
- **Macro dependence.** Economic indicators dominate the model, so it partly learns *when* calls were made. The data covers a single unusual period (2008-2010, the financial crisis).
- **Association, not causation.** No experiment was run; effects of channel, month or call count may reflect how the bank chose its targets.
- **Illustrative economics.** Cost and value per call are assumptions.

7. Recommendations and next steps

- **Rank before dialling.** Score customers and work the list top-down, stopping where expected conversion drops below break-even.
- **Re-activate past responders first** (63.8% conversion on 3.7% of the base) and give past non-responders a new offer.
- **Shift to cellular and cap repeat attempts**; conversion falls below 5% after eight calls.
- **Pilot tailored offers** for retirees, students and under-25s, with a randomised holdout to measure true uplift.
- **Validate and monitor:** retrain regularly, run a live holdout before scaling, and replace assumed costs with real figures.